

COMP2005

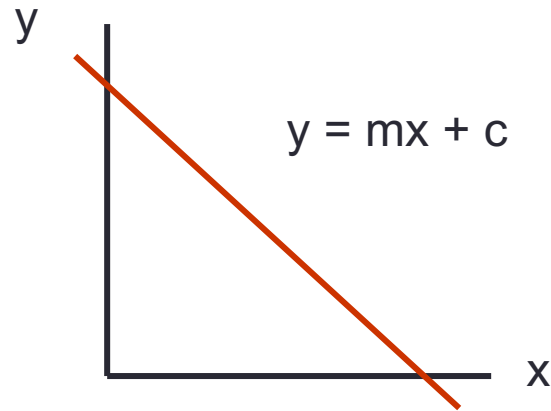
The Hough Transform

Line Parameters

- The standard equation of a line is

$$y = mx + c$$

- If we know the line parameters m, c we can vary x , compute y and draw the line
- This line represents the set of (x, y) pairs that satisfy the above equation



But we don't know the line parameters - we just have some data points (x_i, y_i)

Parameter Space

- But if a data point (x_i, y_i) lies on a line then that line's parameters m, c must satisfy

$$y_i = mx_i + c$$

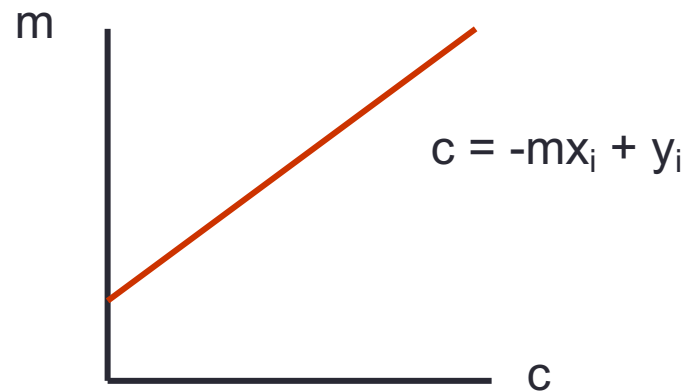
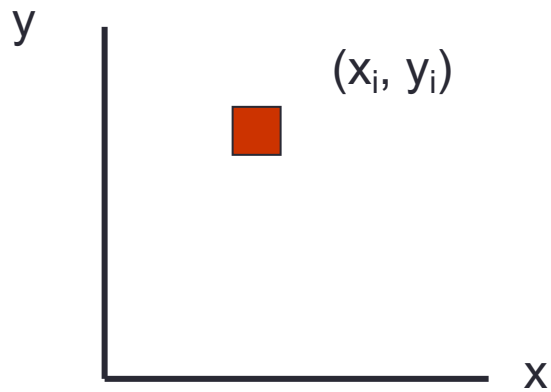
- So we can represent all possible lines through (x_i, y_i) by the set of (m, c) pairs that satisfy this equation

- Rearranging the equation gives

$$c = -mx_i + y_i$$

- This also describes a straight line, but as x_i and y_i are known and m and c are unknown that line is in m, c space – a parameter space

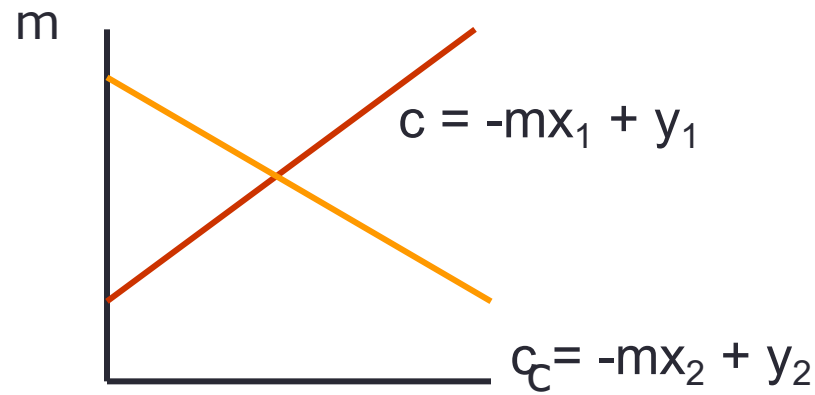
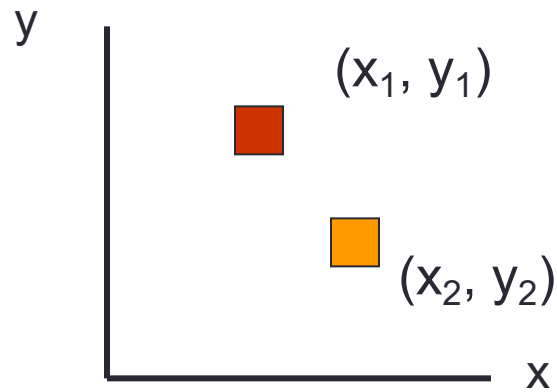
Parameter Space



- This straight line in m, c space represents all the values of (m, c) that satisfy $y_i = mx_i + c$, and so **the parameters of all the lines that could pass through (x_i, y_i)**

Parameter Space

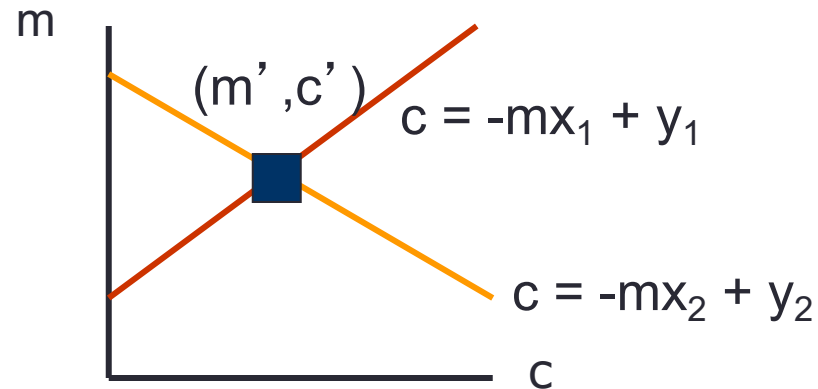
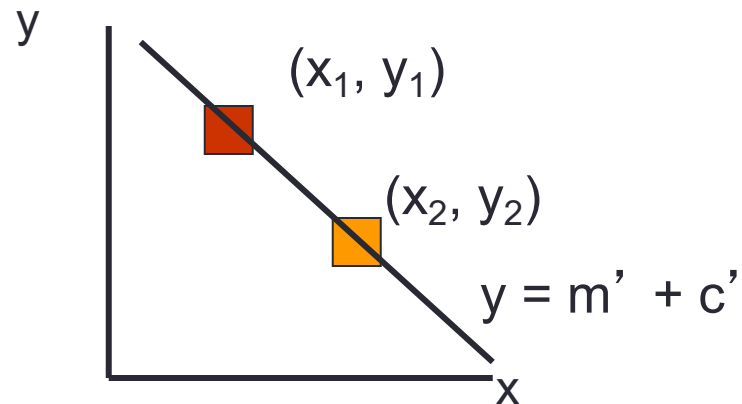
- Suppose there are 2 edges in our image (x_1, y_1) and (x_2, y_2)



- They will each generate a line in (m, c) space representing the set of lines that could pass through them

Parameter Space

- The intersection of our 2 lines is the intersection of 2 sets of line parameters



- The point of intersection therefore gives the parameters of a line through both (x_1, y_1) and (x_2, y_2)

Parameter Space

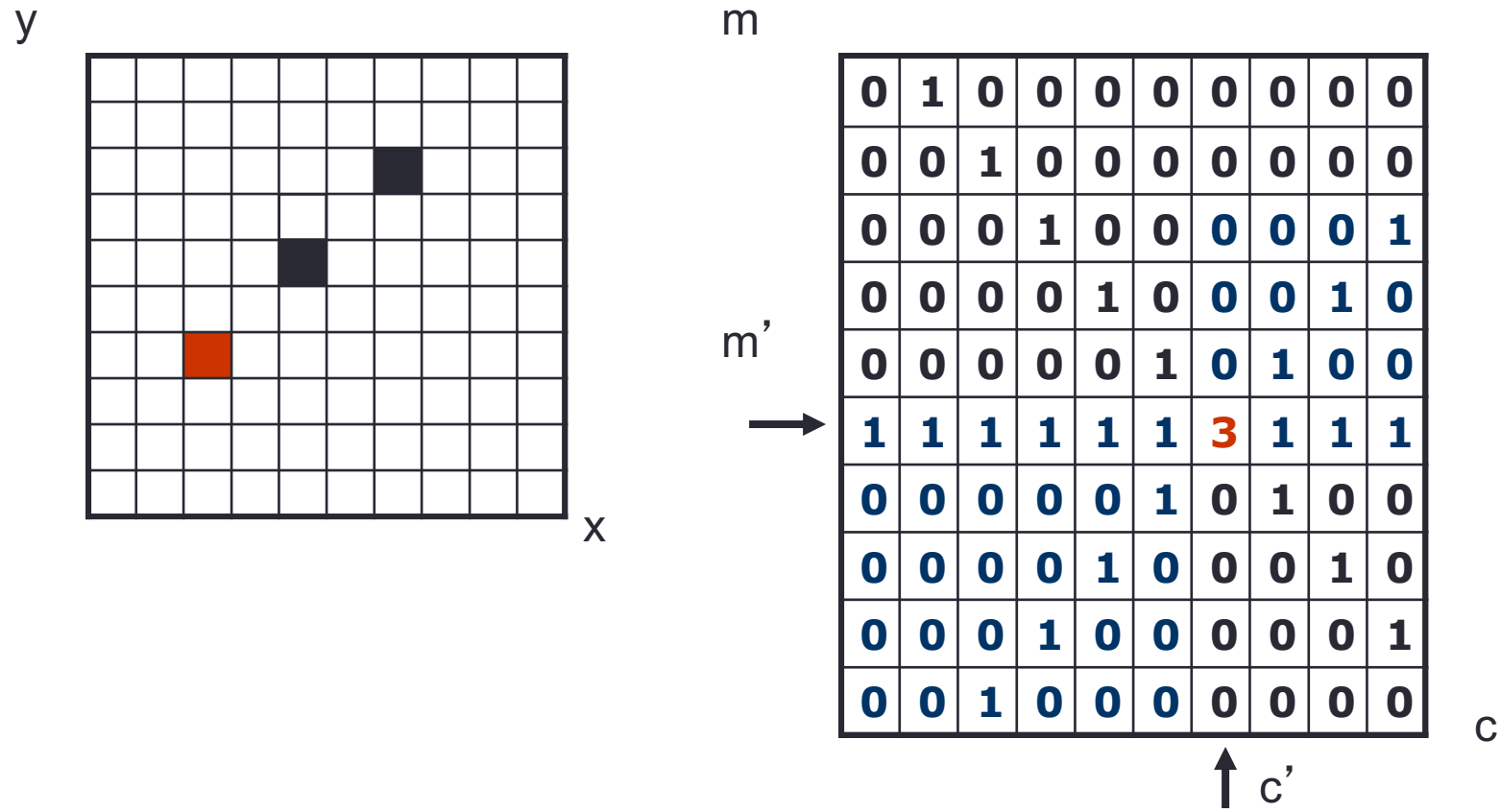
- Taking this one step further
 - All pixels which lie on a line in (x,y) space are represented by lines which all pass through a single point in (m,c) space
 - The point through which they all pass gives the values of m' and c' in the equation of the line: $y=m'x+c'$.
- To detect lines all we need to do is transform each edge point into m,c space and look for places where lots of lines intersect

This is what Hough Transforms do

The Hough Transform

1. Quantise (m,c) space into a two-dimensional array A for appropriate steps of m and c
2. Initialise all elements of $A(m,c)$ to zero
3. For each pixel (x_i, y_i) which lies on some edge in the image, we add 1 to all elements of $A(m,c)$ whose indices m and c satisfy $y_i = mx_i + c$
4. Search for elements of $A(m,c)$ which have large values - each one found corresponds to a line in the original image

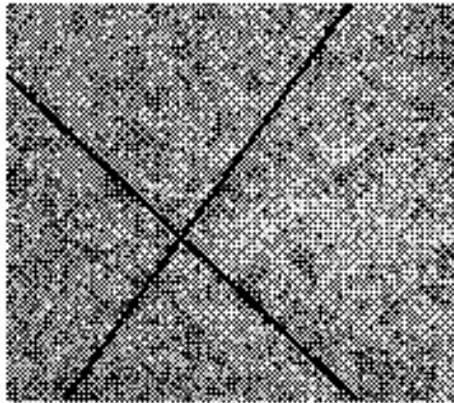
The Hough Transform



There is one line: $y = m' x + c'$

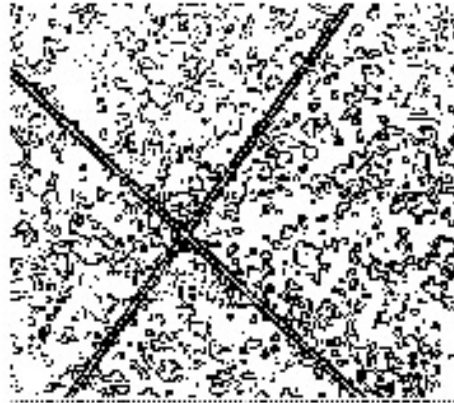
A (More) Real Example

Image



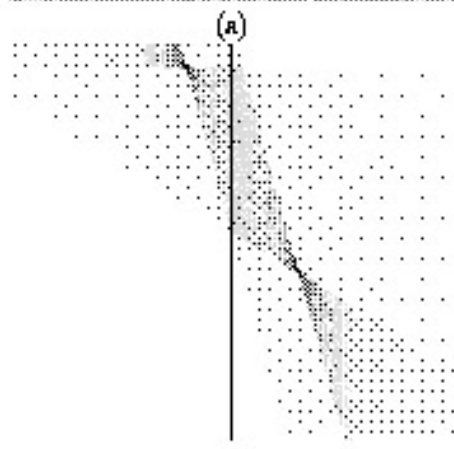
(a)

Edges



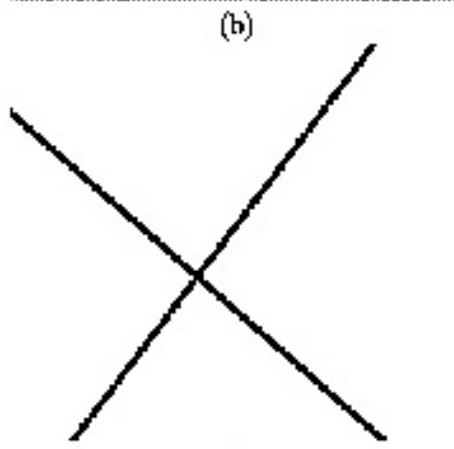
(b)

Accumulators



(c)

Result



(d)

Line Parameters

- The Hough Transform only supplies the parameters of the lines it detects; this can be an advantage or a disadvantage

